

Indian Institute of Technology (BHU), Varanasi
Department of Mathematical Sciences
Mid Semester Examination

Course Title: Numerical Techniques

Course Code: MA 201

Time: 2 Hours

Note: Scientific calculators are allowed.

Session: 2023-24

Semester: Even

Maximum Marks: 30

Q. 1.

(a). Using 4-digit arithmetic with rounding, solve the system of equations $Ax = b$ by Gauss elimination method with partial pivoting, where

$$A = \begin{pmatrix} -3.5 & -7.4 & -0.1 & 3.7 \\ 4.0 & -2.0 & 1.5 & -6.4 \\ 5.0 & 2.0 & 3.0 & -1.0 \\ 4.5 & 4.2 & 8.9 & -4.1 \end{pmatrix}, \quad b = \begin{pmatrix} 22.3 \\ -9.7 \\ 1.0 \\ -7.3 \end{pmatrix}.$$

(b). If a matrix is diagonally dominant, is there ever any need for partial pivoting during the Gaussian elimination procedure?

[5]

Q. 2. Given

$$A = \begin{pmatrix} 3/2 & 1/2 \\ 1/2 & 3/2 \end{pmatrix}.$$

For what values of α does the vector sequence $\{y_n\}_{n=0}^{\infty}$ defined by

$$y_n = (I + \alpha A + \alpha^2 A^2) y_{n-1}, \quad n = 1, 2, \dots$$

y_0 arbitrary, converges to 0 as $n \rightarrow \infty$?

[5]

Q. 3. Assume $g(x)$ and $g'(x)$ exist and are continuous on $[a, b]$; and further, assume $a \leq x \leq b \Rightarrow a \leq g(x) \leq b$, and $\lambda \equiv \max_{a \leq x \leq b} |g'(x)| < 1$. Let α be the unique solution of $x = g(x)$.

Then, show that

(i). The equation $x = g(x)$ has a unique solution α in $[a, b]$.

(ii). For any initial guess x_0 in $[a, b]$, the iteration scheme $x_{n+1} = g(x_n)$, $n = 0, 1, 2, \dots$ will converge to α .

[2.5+2.5]

Q. 4. Use Newton method with 5-digit arithmetic rounding to solve the equation $x^3 = 0.165x^2 - 3.993 \times 10^{-4}$ taking an initial guess as 0.05. Perform three iterations of the method. Also, calculate the absolute relative approximate error at the end of each iteration.

[5]

P.T.O.

Q. 5. Suppose a table is to be prepared for the function $f(x) = e^x$, for $x \in [0, 1]$. What step size h will ensure that linear interpolation gives an absolute error of at most 10^{-6} for all $x \in [0, 1]$? You are allowed to assume the standard interpolation error formula. Give complete details to get full marks.

[5]

Q. 6. Use Newton divided difference formula with 5-digit arithmetic rounding to calculate $f(0.25)$ from the following table

x	0.1	0.2	0.3	0.4	0.5
$f(x)$	9.9833	4.9667	3.2836	2.4339	1.9177

[5]